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Task Description:

The objective of this research was to investigate the fundamental physics of films undergoing condensation in normal- and reduced gravity. Of particular interest were the growth in film thickness, the film stability, interfacial motion, and heat transfer. The primary test configuration was a condensing film on the underside of a flat, cooled surface. Experimental tests, combined with numerical modeling, were employed to gain an improved understanding, within the limitations of a ground-based experiment, of the fundamental fluid physics responsible for condensate film growth and film stability, and the corresponding implications for heat transfer.
The experimental effort involved two condensing film configurations using similar test cells. First, condensation tests were conducted in the laboratory with a stable condensing film growing on an upwards-facing, horizontal condensing surface. These tests primarily allowed the validation of experimental hardware and diagnostics. Subsequently, the test cell was inverted to study the development of fluid instabilities due to body force with and without the presence of condensation. The presence of a body force directed away from the wall due to gravity (at normal or a reduced level) can drive the Rayleigh-Taylor instability, resulting in surface waves and the eventual formation of pendant drops. Determining the features of competition between the destabilizing force and the stabilizing surface tension forces was one objective of the proposed work. Both laboratory experiments involved a continuous vapor supply. A somewhat different configuration was developed for aircraft experiments for reduced-gravity tests to be flown on board parabolic-trajectory aircraft. In this case there was a closed test cell with pressure control and a constant-pressure expansion chamber.

Two fundamentally different types of fluids were employed in this research to help isolate the fluid mechanical and thermal mechanisms operative in condensing films. These were a) a condensing film and b) a non-condensing film with film growth provided by mass addition through a perforated plate. The first fluid environment had all features associated with film condensation, including mass addition and heat transfer effects. The latter configuration allowed the study of the effects of film growth on layer stability, but without thermal effects. Also planned was the study of a cooled, non-condensing film in which case thermocapillary effects could be present in reduced gravity owing to the cooling of the film, but without the effects of mass addition and film growth.

Condensation experiments were conducted in normal gravity in a test cell containing a cooled brass block with an exposed diameter of 10 cm. Vapor was provided by evaporating n-pentane in a 500 ml degassed Pyrex container. N-pentane was selected as the primary working fluid due to its excellent wettability characteristics and its high vapor pressure (boiling point 36 C). The temperature at the exposed surface of the brass block was regulated by a circulating temperature-controlled water/glycol mixture. A vacuum pump ensured positive flow of vapor through the test cell during the course of each experiment. The diameter of test cell was sufficiently large to minimize the impact of edge effects at the center of the surface on which condensation occurs.

Film behavior under unsteady conditions was examined by imposing a cyclic variation on the system pressure to examine the full range of condensing film behavior, ranging from thin films, unstable films, pendant drops, and subsequently film evaporation leading to film dry-out. The pressure of the vapor was controlled by a closed-loop pressure control system. Validation experiments were conducted for unstable (i.e. surface facing downwards) condensing films under steady-state conditions for system pressure ranging from 60 kPa to 75 kPa and subcooling of up to 5 C. The measured heat transfer under steady-state conditions resulted in Nusselt numbers in agreement with previously published correlations for steady, unstable films.

In addition to the condensing films discussed above, films of time-varying thickness but in the absence of phase change were examined experimentally. In this case film growth was provided by mass addition through a perforated plate to simulate the growth of a condensing film. The non-condensing cell consisted of a stack of thin stainless steel disks 10 cm in diameter mounted in a brass enclosure. The disks were perforated with a regular pattern of 361 holes each 0.25 mm in diameter. The non-condensing experiments were conducted in -1g (surface facing downwards) using 50 cSt and 125 cSt silicone oil pumped through the perforated disks at a specified rate by a syringe micropump.

Polishing the metal surface of the cooled plate allowed for double-pass shadowgraph imaging. The shadowgraph imaging was performed to determine the features of film instability, including the wavelengths, surface deformation, and droplet formation and departure rates. Images were acquired by a CCD camera at video rates and are synchronized with the temperature and heat transfer measurements. The shadowgraph system was also used in the imaging of the non-condensing films.

The test surface was instrumented with a heat transfer gauge and thermocouples. The embedded thin film heat flux sensor measured the heat flux and temperature without disturbing the fluid film. The sensor was ringed by a film of Kapton of identical thickness and thermal conductivity as the heat flux gauge to ensure uniform thermal resistance across the cross section of the test chamber. Thermal data were acquired using a PC-based system, which also acquired the video images of the film to allow for synchronization of the shadowgraph imaging with the temperature and heat flux measurements. A one-dimensional, finite-difference, inverse heat conduction algorithm was developed to calculate the unsteady heat flux at the surface using data acquired by the embedded heat transfer sensor. The numerical method divided the heated/cooled block into two regions, a direct region between the sensor location and the heat source/sink plane (the circulating liquid) and an indirect region between the sensor location and the exposed surface. The solution for the temperature and heat flux in the direct region solution was straightforward. An inverse method was used in the indirect region. The heat flux input to the algorithm used a non-sinusoidal curve fit to heat flux data over a typical condensation/evaporation cycle. The accuracy of the inverse method was verified by comparison with known temperatures and heat fluxes for the case of a dry surface.

Ultrasonic measurements of the film thickness were performed for both condensing and non-condensing films. In the first realization of the system, a linear array of eight Panametrics Model M110-5 transducers of 6 mm diameter were pulsed with a square-wave signal at a frequency of 5 MHz and a pulse duration of approximately 0.3 ms. The transducers were excited by a Parametrics Model 5077PR Pulser/Receiver unit, which also received the detected echo signals, which were then amplified and sent to the PC laptop. For thin films (60 μm to 2-3 mm thick) the layer thickness was determined by frequency analysis. For thicker layers, time-domain analysis was performed to generate directly the layer thickness. The accuracy of this technique was verified using both static layers and non-condensing layers with known rate of fluid addition. Numerical modeling using Wave2000 software by Cyber Logic, Inc. was employed to verify the experimental operation and to predict the outcome of measurements under various test conditions. An improved ultrasound measurement system consists of a Panametrics-NDT model 25MX Plus pulser/receiver generator combined with a GE Panametrics model MX8 multiplexer. The three transducers are Panametrics-NDT models M110-RM (6 mm diameter, 5.0 MHz), M112-RM (6 mm diameter 10.0 MHz), and M116-RM (3 mm diameter 20.0 MHz). The improved system allows for the measurement of film thicknesses as small as 50 to 100 μm, with a resolution

of roughly 10 to 20 μm and measurement times on the order of 10 μs.

An optical imaging system based on the Michelson interferometer was developed to provide a map of the surface features of the condensing films. Both monochromatic laser light and white light sources were employed. Laser light was provided by a B&W Tek BWT-5E Nd:YVO thermoelectrically cooled, diode pumped frequency doubled to give a wavelength of 532 nm; white light was generated by an LED-based fiber optic system. In both cases a Ronchi ruling was employed to obtain the necessary fringe pattern. Images of the deformed fringes on the copper block surface were captured by a CCD camera and analyzed on a PC laptop using FringeProcessor software. These images were processed using Fourier transform analysis. Tests of the interferometer system were first performed by placing a droplet of water on the test surface from which its surface map was successfully reconstructed. The droplet tests gave an estimated out-of-plane resolution of approximately 25 μm and a nominal in-plane resolution of approximately 1 mm. Test with condensation were performed using a condensation cell that consisted of a Pyrex cylinder 170 mm high and 100 mm in diameter.

Numerical modeling of condensing films involved the solution of the unsteady Navier-Stokes equations by the Finite Element Method based on a front tracking technique with a moving and deformable mesh. The code considered condensation of the film under the influence of gravity, and surface tension with periodic boundary conditions. Initial simulations employed a two-dimensional rectangular domain, which deforms with time. The rate of condensation is related to the rate of energy conducted to the surface and the heat exchanged by convection. The program simulated the film behavior modifications that occur due to the action of condensation. The liquid film growth leading to Rayleigh-Taylor instability in isothermal, non-condensing films was also modeled numerically. In this case the Navier-Stokes equations were solved by a conventional Finite Volume Method in two dimensions. The moving fluid interface was treated as a moving internal boundary connected to marker points in separate grids aligned with the interface to calculate the fluid motion on either side of the front. The motion of the fluid interface was calculated as a part of the solution, using the appropriate boundary conditions across the interface.

A test rig was developed to support condensation experiments in reduced gravity on board NASA's parabolic-trajectory aircraft. Rapid heat transfer is achieved by several thermoelectric coolers in conjunction with a pressure/volume control system. The test cell is a 12 cm diameter, 25 cm long glass cylinder with copper plates at the bottom and at the side. The surface on which the evaporating/condensing films reside is 2.5 cm thick to maintain a nearly constant wall temperature. The other plate (0.64 cm thick) is rapidly heated or cooled to provide a vapor source/sink as needed to minimize the required volume change during film evaporation and condensation. The pressure/volume control system consists of an expansion chamber fitted with a bladder connected to the test cell and a pressure control system which regulates the pressure in the expansion chamber outside of the bladder (and hence in the test cell). The system is capable of quickly achieving and maintaining the desired system pressure required to initiate evaporation and condensation under reduced-gravity conditions. The test chamber allows for optical access to the evaporating/condensing film using the same optical configuration employed for imaging in the normal-gravity experiments. The copper surface on which the evaporating/condensing films reside is instrumented with temperature and heat transfer sensors, or, alternatively, ultrasound transducers for film thickness measurement.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Potential for improving heat transfer and condensation rate in liquid/vapor systems in space applications. Demonstrations of the use of ultrasound gauging to measure film thickness and growth rate under dynamic conditions in condensing and non-condensing films and for other fluid processes.

Task Progress:

Tests were performed in normal gravity for both stable and unstable films, with and without condensation. For the case without condensation a pumped film was employed to simulate layer growth in the absence of thermal and/or phase change effects. Specific conclusions and results include:
• For unsteady condensing films, the heat flux and the surface temperature vary substantially over the course of each cycle. Heat flux increases most dramatically due to the initial formation of a thin film of condensate. The spatially-averaged heat flux also appears to be rise significantly at the appearance of the Rayleigh-Taylor instability and drop formation during pressure ramp-up. For sufficiently long time, a steady value of heat transfer occurs, as expected, accompanied by continuous droplet formation and break-off.

• For cycles with the lowest difference between the vapor and surface temperatures, increases in heat flux due to natural convection prior to the development of the condensate film are observed. For larger temperature differences this phenomenon is less apparent, being quickly overwhelmed by the onset of condensation.

• For the case of cyclic pressure variation a degree of hysteresis occurs, with the heat flux and surface temperatures being different during the evaporation and condensation portions of the cycle for a given system pressure. The heat flux for these unsteady conditions thus appears to be influenced by more than the degree of superheating of the film.

• Inverse methods can be implemented to determine cyclic surface conditions with data from an embedded probe in steady and unsteady experiments involving film condensation. Systematic examination of the effect of the location of the embedded heat transfer sensor on the estimated surface heat flux suggest a critical depth above which the sensor position can cause significant errors in the calculated surface heat flux and surface temperature.

• Non-condensing, growing-film experiments showed that the time to pendant droplet formation and break-off decreases monotonically as the pumping rate is increased. The film thickness at the point of first droplet break-off increases also with increased pumping rate. No significant change in the instability wavelength with pumping rates was observed. Furthermore, the increased film thickness at the onset of instability suggests that mass addition by condensation has a stabilizing effect on the Rayleigh-Taylor instability.

• The applicability of the ultrasound technique for thickness measurement of fluid layers with thicknesses down to a fraction of one wavelength was successfully demonstrated for static films and for condensing films on the underside of a cooled plate. In the latter case, oscillations in film thickness due to the passage and/or shedding of droplets from the cooled plate surface were observed. The ultrasound technique was also successfully employed to measure the wavelengths and wave propagation speeds of surface waves driven by an external source.

	• Applying a numerical acoustic model to non-flat films indicated a critical inclination of the layer surface beyond which the ultrasound method can cease to provide accurate film thickness measurements. The modeling also indicated that the ability of the ultrasound method to detect the presence of pendant drops is strongly dependent on drop position. • Interferometric techniques were seen to be capable of accurately rendering the surface contours of simple fluid configurations, specifically water drops on a solid surface in open air. Interferometric imaging of condensing films using were less successful largely owing to distortions in the fringe pattern by the condensation test cell. • Numerical finite element modeling was successfully conducted for the case of condensing films in zero gravity. The results confirmed the stability regions expected under different terrestrial conditions but results were not conclusive for the cases of microgravity conditions. Finite volume modeling of the interfacial behavior of non-condensing films indicated changes in film stability with pumping rate qualitatively similar to those observed in the non-condensing, growing film experiments. Thus film stability is seen to impact on the interfacial behavior and heat transfer rate of condensing films. Also of interest is the hysteresis phenomenon -- that changes in the film thickness and the rate of heat transfer are not uniquely determined by the degree of superheat for unsteady films with phase change, but depend also on the portion of the heating/cooling cycle. This may have important implications for the start-up and shut-down behavior of power generation and other spacecraft systems involving condensation. The development of the ultrasound technique, optical interferometry, and heat transfer measurements using corrected, embedded probes may also prove useful in the further study of films with condensation.
Bibliography Type:	Description: (Last Updated: 09/03/2025)
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